

Aditya

LLM Systems Engineer | Agentic AI | Context-Oriented Architect



New Delhi, India



Adyypower



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PROJECTS

CONTEXT HYGIENE SYSTEM | LANGGRAPH, AGENTIC AI, CONTEXT CONTROL

Jan 2026

- Designed a context hygiene layer for agentic systems to control long-term conversation memory and token growth.
- Implemented adaptive context pruning, history compression, and token-threshold monitoring to prevent context window overflow.
- Added drift detection and state-transition mechanisms using LangGraph to maintain structured conversational flow.
- Improved reasoning stability in multi-step agent workflows by preserving task-critical context while filtering noise.

ADVANCED ADAPTIVE RAG SYSTEM | LANGCHAIN, CHROMADB, WEB SCRAPING, MULTI-LLM

SUPPORT

oct-nov 2025

- Designed a multi-source ingestion pipeline supporting PDF uploads and dynamic web scraping, storing embeddings in ChromaDB.
- Implemented hybrid retrieval (semantic + keyword-based) with LLM-based document grading for relevance filtering.
- Built a self-evaluating generation loop that classifies outputs as useful, not useful, or unsupported.
- Integrated fallback web search and regeneration mechanisms to improve answer completeness.
- Architected modular LLM abstraction layer enabling seamless model switching.
- Reduced unsupported responses by introducing conditional fallback retrieval based on LLM output grading.

ADVANCED RAG WITH LCEL & LOCAL LLM | LANGCHAIN LCEL, LOCAL LLMs, FAISS

sep-oct 2026

- Built a fully local RAG pipeline using LangChain Expression Language.
- Enabled private document QA without cloud dependency.

MEDIAGENT SYSTEM | MULTI-AGENT AI, LLMs

dec 2025

- Developed a multi-agent medical assistant for structured symptom analysis.
- Coordinated planner and executor agents for controlled reasoning.

GRAMMAFIX – AI GRAMMAR CORRECTION | FLASK, T5, GEMINI API

june 2025

- Built a hybrid NLP web app for real-time grammar correction.
- Integrated local and cloud LLMs for low-latency inference.

LLAMA-2 7B FINE-TUNING | PYTORCH, QLoRA, LoRA, HUGGINGFACE

june 2025

- Fine-tuned LLaMA-2 7B using QLoRA on Google Colab for grammar error reasoning tasks.
- Trained on datasets from HuggingFace and Kaggle using parameter-efficient fine-tuning.

ENGINEERING PERSPECTIVE

- Design controllable autonomous systems around LLMs using structured state-driven workflows and context management.
- Focus on reliability-first architectures that reduce drift and stabilize reasoning.
- Build modular, interpretable agentic systems that balance autonomy with structured control.

SKILLS

PROGRAMMING

Python • C++ • C
SQL • Java

FRAMEWORKS

PyTorch • Hugging Face
LangChain and LangGraph
• Flask

GENERATIVE AI

LLMs
Fine Tuning and Pre Tuning
Prompt Engineering
LoRA • QLoRA

CONCEPTS

Machine Learning
Deep Learning
MAS & Agentic AI
Model Optimization
Agents & RAG Systems

INFORMATION

Languages: English (Fluent), Japanese (Basic)

Cricket: District Level;
DDCA U-16 State Camp

EDUCATION

VIVEKANANDA GLOBAL UNIVERSITY

B.TECH IN COMPUTER SCIENCE &
ENGINEERING

Aug 2023 -- May 2027 | Jaipur, India
Third Year (Current)

JM SARVODAYA VIDYALAYA, SEC-12

HIGHER SECONDARY EDUCATION
2021 | New Delhi, India
Percentage: 85%

CERTIFICATIONS

IBM

Generativ AI engineering
Agentic AI (Coursera)
Machine Learning
(Coursera)

DeepLearning.AI

Deep Learning
Specialization